SIDDHARTHA ACADEMY OF HIGHER EDUCATION

EVALUATION OF COMMUNITY SERVICE INTERNSHIP

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Introduction Intelligent Monitoring and Predicting Airborne Infection Risks in Hospital Environments Using ML-Enabled IoT

AIRBORNE INFECTIONS ARE A MAJOR CAUSE OF HOSPITAL-ACQUIRED INFECTIONS DUE TO POOR VENTILATION AND DEGRADED INDOOR AIR QUALITY. MANUAL MONITORING METHODS ARE INEFFICIENT AND REACTIVE. THIS PROJECT PROPOSES AN IOT-BASED SMART SYSTEM THAT CONTINUOUSLY MONITORS HOSPITAL AIR QUALITY AND PREDICTS AIRBORNE INFECTION RISK LEVELS USING MACHINE LEARNING, ENABLING EARLY WARNINGS AND PREVENTIVE ACTION.

Objectives

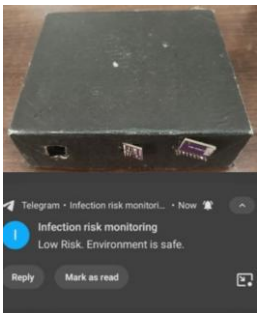
- TO CONTINUOUSLY MONITOR HOSPITAL INDOOR AIR QUALITY USING IOT SENSORS.
- TO CLASSIFY HOSPITAL ZONES INTO LOW, MEDIUM, AND HIGH INFECTION RISK
- TO PROVIDE REAL-TIME ALERTS FOR INFECTION-PRONE CONDITIONS

METHODOLOGY

AN ESP32-WROOM-32 MICROCONTROLLER COLLECTS REAL-TIME DATA FROM ENVIRONMENTAL SENSORS.THE COLLECTED DATA IS PREPROCESSED AND ANALYZED USING K-MEANS++ CLUSTERING TO CLASSIFY INFECTION RISK LEVELS. RESULTS ARE UPLOADED TO THE THINGSPEAK CLOUD, WHERE ALERTS ARE TRIGGERED.

RESULT

THE SYSTEM ACCURATELY DETECTS AND CLASSIFIES AIRBORNE INFECTION RISK LEVELS IN HOSPITAL ENVIRONMENTS AND GENERATES REAL-TIME ALERTS .



CONCLUSION

AN ESP32-BASED IOT SYSTEM WITH K-MEANS CLUSTERING ENABLES ACCURATE REAL-TIME AIRBORNE INFECTION RISK PREDICTION IN HOSPITALS.

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Video Demonstration: